Haotian Fu

haotian.fu01@gmail.com | Website | LinkedIn | (+1) 9342559282 | Stony Brook, NY

EDUCATION

Stony Brook University **Sep 2024 - June 2029 (Expected)**
Ph.D. in Computer Engineering. Advisor: Prof. Fan Ye, IEEE Fellow
Research Interests: Multimodal Sensing, Sensing Infrastructure, Spatial Intelligence.
Shanghai Jiao Tong University (SJTU) **Sep 2020 - Aug 2024**
B.E. in Electrical and Computer Engineering (ECE) 2024 Outstanding Graduate of SJTU
Minors: Computer Science (CS), Data Science (DS)

RESEARCH PROJECTS

Graduate Research Assistant | Stony Brook University

Multimodal Dataset Collection, Curation, and Quality Assessment **Mar 2025 - present**

- Led data acquisition and curation for **113 long-duration sessions** in a four-room apartment sensing dataset, integrating **16 UWB sensors** on **16 Raspberry Pis**, **6 Kinect video/depth cameras**, and **3 wearable watches/fingertips** to capture continuous activities of daily living (ADLs).
- Developed and maintained an **Android application** and **structured data collection protocol**, incorporating participant guidance, timestamped activity annotations, and UI/UX design to support consistent, repeatable and scalable large dataset collections.
- Implemented and tested **an automated data quality assessment and metadata acquisition workflow**, combining session- and modality-level integrity checks on availability, completeness, and consistency.
- Mentored undergrad/master's students in data collection, metadata acquisition, quality assessment, and Android application development, coordinating contributions to ongoing dataset formulation.

Multi-Sensor Fusion for Robust Indoor Human Localization and Tracking **Nov 2024 - Feb 2026**

- Identified range estimation as a critical bottleneck in realistic indoor localization, revealing failure patterns caused by multipath reflections, occlusion, and environmental clutter during continuous ADLs.
- Developed **ScoreKFT**, a reliable and interpretable physics-informed range estimation and particle-filter-based multi-sensor fusion framework integrating **16 UWB sensors**, achieving **0.958 m range RMSE** and **over 30% lower error** than representative baselines.
- Developed a **DBSCAN-based ground-truth generation pipeline** using Kinect video/depth data to reconstruct human locations and trajectories without predefined movement paths.
- Validated localization and tracking performance across 12 long-duration ADL sessions in a cluttered, fully furnished four-room apartment, establishing a **cross-modal evaluation methodology and benchmark** for realistic indoor home sensing.

Cross-Population Generalization Evaluation for ADL Recognition (ADL-CLIP) **July 2026 - present**

- Reproduced a Transformer-based RF–text contrastive learning framework with PELT-based temporal segmentation and iterative representation refinement for continuous ADL detection.
- Conducted cross-population training and evaluation experiments for the ongoing **journal extension**, using **Slurm-managed NVIDIA H200/B40 GPU clusters**, and troubleshooting PyTorch/CUDA environments to assess model generalization across participant populations.

PUBLICATION

H. Fu, M. Liu, H. Han, Z. Xie, and F. Ye, "ScoreKFT: Robust Scoring-Based Range Estimation for Human Tracking in Realistic Home-Like Environments with UWB Radars," *2026 IEEE Global Communications Conference*, Accepted.

H. Liu, **H. Fu**, S. Yuan, C. Wu, Y. Luo and J. Li, "Adaptive Processing for Video Streaming with Energy Constraint: A Multi-Agent Reinforcement Learning Method," *2023 IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, 2023, pp. 122-127, doi: 10.1109/GLOBECOM54140.2023.10437053.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, R
ML & Scientific Computing: PyTorch, CUDA, NumPy, SciPy, scikit-learn
Infrastructure & Tools: Git, Slurm, Docker, Linux, AWS (S3, Lambda)